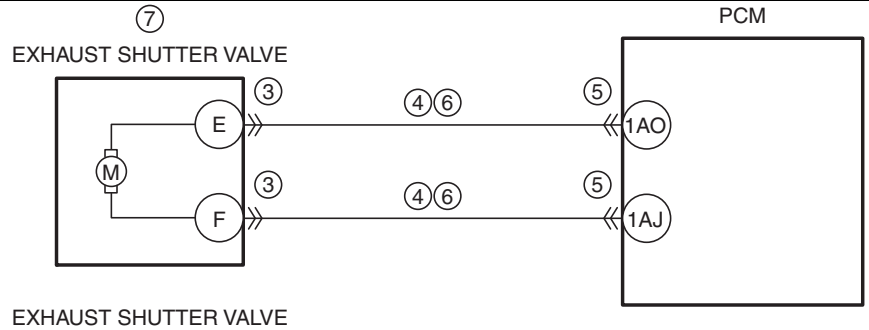
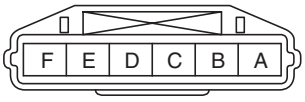
DTC P0475:00 [SKYACTIV-D 1.5]

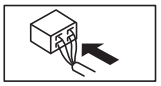
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DTC P0475:00	Exhaust shutter valve DC motor control circuit problem
DETECTION CONDITION	• The exhaust shutter valve drive current exceeds the specification or the exhaust shutter valve drive circuit temperature is 175 °C {347 °F} or more for a continuous **3 s** when the following condition is met. **MONITORING CONDITIONS** — The following DTCs are not detected: • PCM internal CPU: P0606:00 **Diagnostic support note** • This is a continuous monitor (EGR system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the EGR control • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Exhaust shutter valve connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — Exhaust shutter valve terminal E—PCM terminal 1AO — Exhaust shutter valve terminal F—PCM terminal 1AJ • PCM connector or terminals malfunction • Short to power supply in wiring harness between the following terminals: — Exhaust shutter valve terminal E—PCM terminal 1AO — Exhaust shutter valve terminal F—PCM terminal 1AJ • Exhaust shutter valve malfunction • PCM malfunction

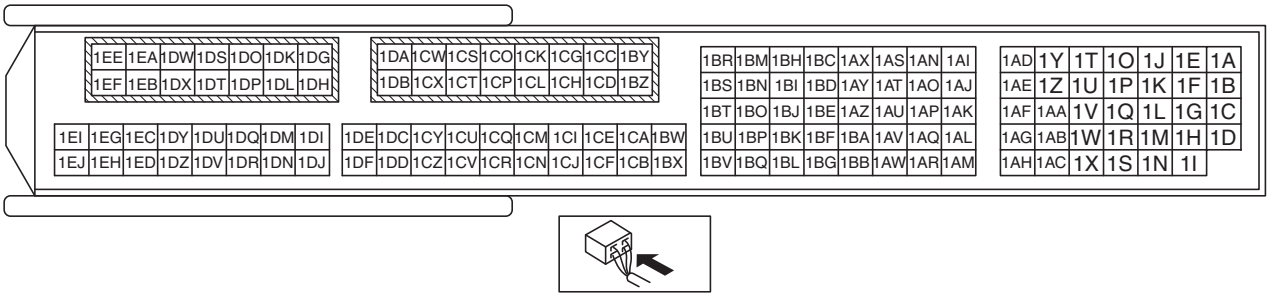


EXHAUST SHUTTER VALVE WIRING HARNESS-SIDE CONNECTOR





PCM WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD FREEZE FRAME DATA/SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (A/F sensor related) on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT EXHAUST SHUTTER VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the exhaust shutter valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.
4	**INSPECT EXHAUST SHUTTER VALVE CIRCUIT FOR SHORT TO GROUND** • Verify that the exhaust shutter valve connector is disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Exhaust shutter valve terminal E — Exhaust shutter valve terminal F • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — Exhaust shutter valve terminal E—PCM terminal 1AO — Exhaust shutter valve terminal F—PCM terminal 1AJ **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 8.
		No	Go to the next step.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.

STEP	INSPECTION		ACTION
6	**INSPECT EXHAUST SHUTTER VALVE CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the exhaust shutter valve and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the following terminals (wiring harness-side): — Exhaust shutter valve terminal E — Exhaust shutter valve terminal F • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Exhaust shutter valve terminal E—PCM terminal 1AO • Exhaust shutter valve terminal F—PCM terminal 1AJ **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 8.
7	**INSPECT EXHAUST SHUTTER VALVE** • Switch the ignition off. • Reconnect all disconnected connectors. • Inspect the exhaust shutter valve. (See EXHAUST SHUTTER VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to the next step. (See EXHAUST SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See EXHAUST SHUTTER VALVE ACTUATOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
8	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
9	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.